



King Abdul Aziz University

Faculty of Computing & Info Tech.

Computer Science Department

Final Lab. Exam (19-11-1444 / 8 - 6 -2023)

Course: CPCS204 (Data Structures II)

Term: Third 2022-2023, Time: 100 Minutes

Student ID: -----Section: ----- Lab# -----

Student Name: -----computer number:-----

Lab Instructor Name: -----

Max. Marks	Obtained Marks
100	

Instructions:

- Sign in the computer using the lab account (To be provided).
- Open the blackboard (BB) and download the exam files.
- Create a project folder with name "Final204_Name_Id_Section" and copy the exam files in it.
- After solving the exam problems zip your folder and upload it an BB.

Exam Problem

Write a JAVA program to check whether a number (entered by the user) is a special number or not. Special numbers have the same value if flipped. For example: **42324**, **524425** are special numbers whereas **5245** is not. You will need to use 2 queues (implemented as arrays) and 1 stack (implemented as linked list).

The algorithms is as follows:

- 1- read a number long or int.
- 2- extract the digits and at the same time add them to one queue and push them in the stack.
- 3- pop the stack contents in the second queue.
- 4- deque both queues at the same time and check if values are equal.
- 5- if the queues are emptied and the values are the same, then the number is a special number. Otherwise, it's not.

Sample Outputs

Enter a number: 524425 The number is special.	Enter a number: 52425 The number is special.	Enter a number: 77 The number is special.
Enter a number: 5 The number is special.	Enter a number: 5425 The number is not special.	Enter a number: 84 The number is not special.

Important Notes

Note that a skeleton program is given to you where most basic parts of the classes are available and run correctly. You only need to do the following:

Complete the missing parts in Main Class as follows:

- 1- Create one stack object of a linked list type.
- 2- Create two queue objects of an array type, with length of 20.
- 3- Write the following methods:
 - a. **extractDigit**: to extract the digits of the number and store them in the first queue and push them into the stack.
 - b. **getFlipped**: to fill the second queue by popping the stack.
 - c. **isSpecial**: to dequeue both queues and check if number is special.

Good Luck ☺